

Lab: Thought Experiment — The Dark Room and the Nature of Cognition

Objective

Explore the philosophical implications of treating cognition as inference through the lens of the dark room problem, precision dynamics, and the relationship between cognition and consciousness.

Part 1: The Dark Room Challenge

Goal: Engage directly with the most famous objection to Active Inference.

"If an Active Inference agent minimizes surprise, it should seek out the most predictable environment — a dark, silent, empty room — and remain there forever."

- State this objection in your own words. Why is it considered powerful?
- Present the standard Active Inference response (prior preferences / C vector).
- Evaluate: Does the response fully resolve the problem, or does it merely push it back one step (i.e., where do prior preferences come from)?

Write your response here

Part 2: Precision and the Experience of Attention

Goal: Connect the formal concept of precision to the phenomenology of attention.

Perform a brief phenomenological exercise: Focus intently on a single word on this page for 30 seconds, then let your attention relax to the room around you.

- Describe the phenomenological difference between focused and relaxed attention.
- How does the concept of precision (up-weighting certain prediction errors, down-weighting others) capture this experience?
- Is precision "the same thing as" attention, or merely an analogy? Defend your answer.

Write your response here

Part 3: Beliefs Without Consciousness

Goal: Probe the relationship between belief and awareness.

Consider blindsight — a neurological condition in which patients deny seeing stimuli in their blind field but can nevertheless "guess" their location far above chance.

- Does the blindsight patient have a "belief" about the stimulus location in the Active Inference sense (a posterior distribution)?
- If yes, this is a belief without conscious awareness. What does this imply about the nature of belief?
- Is it philosophically coherent to attribute beliefs to unconscious systems (thermostats, bacteria)?

Write your response here

Part 4: The Self as Inference

Goal: Apply the Active Inference framework reflexively.

If cognition is inference, then the experience of having a "self" might also be an inference — the brain's best model of its own Markov Blanket.

- What would it mean for the self to be a "belief" rather than a fact?
- How does this relate to Buddhist accounts of *anattā* (no-self)?
- Is a self that is "merely" an inference any less real than a self that is a substance?

Write your response here

Part 5: Synthesis and Position Defense

Goal: Develop a philosophical position on the nature of cognition.

Write a 200-word statement on one of the following:

- "The dark room problem shows that Active Inference is incomplete as a theory of cognition."
- "Precision is the bridge between the physics of the brain and the phenomenology of consciousness."
- "If beliefs are probability distributions, there is no hard problem of consciousness — only a hard problem of precision."

Share with a peer and defend against their strongest objection.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Objection analysis	Engaging with the dark room problem as a test of theoretical completeness
2	Phenomenological reflection	Connecting formal precision to the lived experience of attention
3	Conceptual boundary testing	Probing whether beliefs require consciousness
4	Reflexive analysis	Applying the inferential framework to the concept of self
5	Position defense	Articulating and defending a view on cognition under challenge